

HOMework 8

NUMERICAL ANALYSIS

(MATH 461) FALL, 2023

DUE: 11/20/2023

Problem 1. Given $h > 0$, find A, B, and C in the following finite-difference approximation, which should give exact values of $f''(x)$ for polynomials of degree 3, regardless of h :

$$D^2 f(x) = \frac{A f(x+h) + B f(x) + C f(x-h)}{h^2} \approx f''(x).$$

Let $f(x) = e^x$. Using the fact that the above finite-difference approximation gives exact values for polynomials of degree 3 to discuss the leading-order error of the differentiation rule.

Problem 2. Use the method of undetermined coefficients to set up the 5×5 system that would determine a fourth-degree accurate finite difference approximation to $f''(x)$ based on 5 equally spaced points,

$$f''(x) = c_{-2}f(x-2h) + c_{-1}f(x-h) + c_0f(x) + c_1f(x+h) + c_2f(x+2h) + O(h^4).$$

Problem 3. Compute the coefficients c_i for the fourth-degree accurate finite difference approximation to $f''(x)$ in [Problem 2](#) and check that they satisfy the system you determined in your answer to [Problem 2](#).

Problem 4. Test the finite difference formula obtained in [Problem 2](#) and [Problem 3](#) to approximate $f''(1)$ for $f(x) = \sin(2x)$ with the following values (in python, `np.logspace(-1, -4, 13, endpoint=True)`) of h :

0.1000, 0.562, 0.0316, 0.0178, 0.0100, 0.0056, 0.0032, 0.0018, 0.0010, 0.0006, 0.0003, 0.0002, 0.0001

Then, produce a plot (both axes logarithmically-scaled) of the absolute value of the error vs. h . You should observe the predicted accuracy for larger values of h , and worse accuracy for smaller values where rounding error dominates.